

- 4 (a) Two resistors X and Y are connected to a power supply of e.m.f. of 8.0 V with negligible internal resistance. The circuit is earthed at point B between the two resistors as shown in Fig. 4.1.

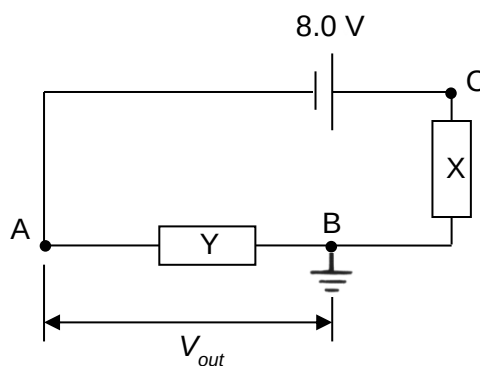


Fig. 4.1

The resistance of Y is $80\ \Omega$.

The current in X is 40 mA.

The potential difference across Y is V_{out} .

- (i) Determine the potential at point A.

potential at point A = V [2]

- (ii) Calculate the resistance of resistor X.

resistance = Ω [2]

- (b)** An NTC thermistor is now connected across resistor Y as shown in Fig. 4.2. The resistance of the thermistor is $100\ \Omega$ when it is at room temperature.

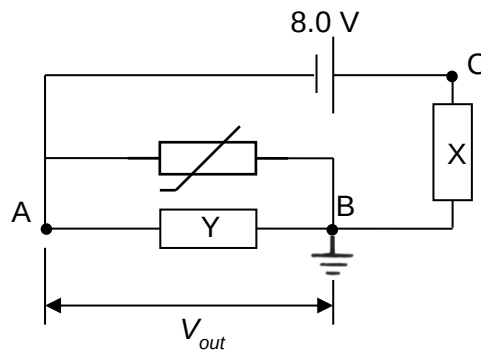


Fig. 4.2

- (i) Calculate V_{out} at room temperature.

$$V_{out} = \dots\dots\dots V \quad [3]$$

- (ii) State and explain whether a fan should be connected across AB or BC so that it turns on when the temperature gets hotter.

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